Lab Procedure for Simulink

Self-Localization

Setup

1. It is recommended that you review Lab 1 – Application Guide before starting this lab.
2. Turn on the QBot Platform by pressing the power button once. To ensure the robot is ready for the lab, check the following conditions.
    a. The LEDs on the robot base should be solid red.
    b. The LCD should display the battery level. It is recommended that the battery level is over 12.5V.
    c. The Logitech F710 joystick's wireless receiver is connected to the QBot Platform. Before use**, always make sure the switch on top is in the X position and that the LED next to the Mode button is off.**
    d. Make sure your computer is connected to the same network that the QBot Platform is on. If using the provided router, the network should be `Quanser_UVS-5G`.
    e. Test connectivity to the QBot. Using the IP displayed in the robot's LCD display, enter the following command in your local computer terminal and hit enter: `ping 192.168.2.x` .
3. Deploy and run qbot_platform_driver_physical on QBot Platform:
    a. Right click on qbot_platform_driver_physical.rt-linux_qbot_platform, select "Show more options", then select "Run on target".
    b. Change Target URI to: `tcpip://192.168.2.x:17000`
    c. Change Model Arguments to `-d /tmp -uri tcpip://192.168.2.x:17099`
    d. Click Run.
    e. The QBot Platform LEDs should pulse white if the driver is deployed and running successfully.
4. Open the Simulink Model localization.slx, as shown in Figure 1. Configure the model so that it can be deployed to the QBot Platform:
    a. Open Hardware Settings under the Hardware ribbon in your model.

b. Expand and browse to `Code Generation > Interface`.

c. Change the MEX-file arguments to the following string including single quotes,

`'-w -d /tmp -uri %u','tcpip://192.168.2.x:17001'`

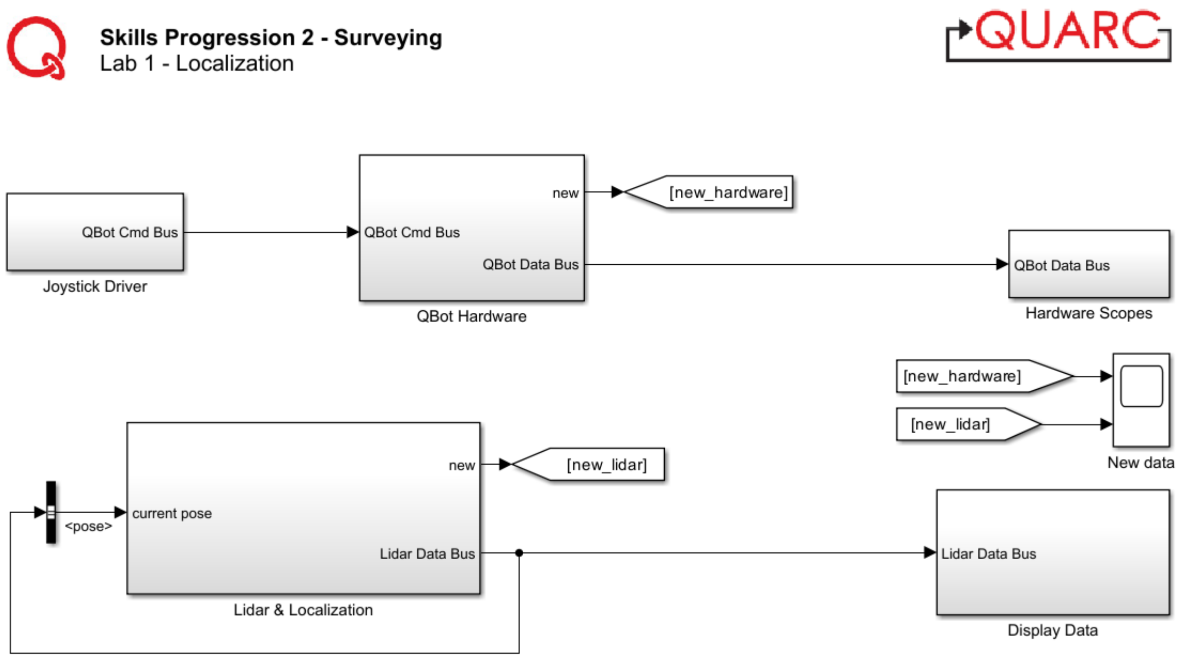

Figure 1: Localization Simulink Model

Introduction to Scan Matching

1. Consider a situation, shown in Figure 2, where the QBot:

    a. Begins at location A
    b. Drives forward
    c. Stops at location B

    Suppose a LiDAR scan is taken at each location, giving you scan A and scan B. Describe how you would relate the data points in scan B to the points in scan A.

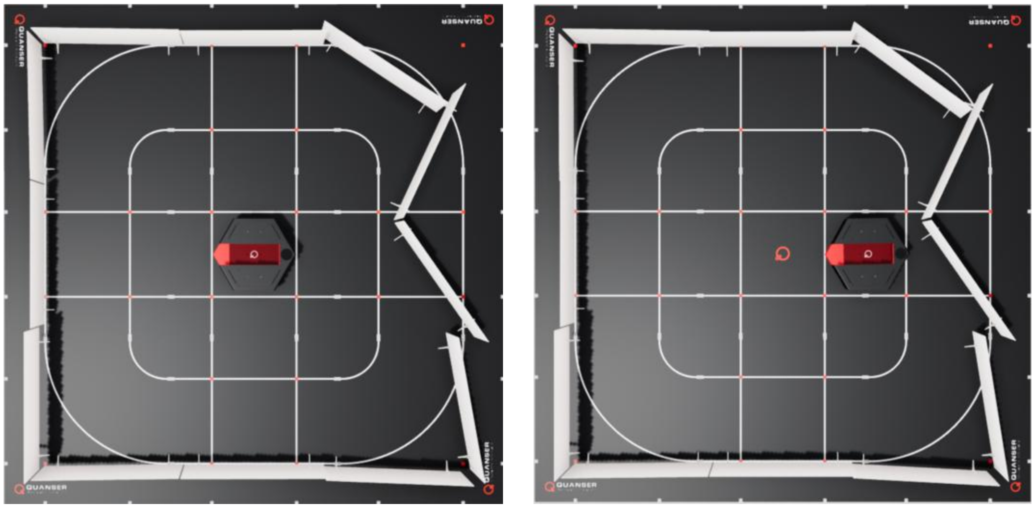

Figure 2. Top view of QBot in location A (left) and location B (right)

2. Consider a situation, shown in Figure 3 below, where the QBot:

    a. Begins facing direction C
    b. Rotates 90 degrees counterclockwise
    c. Stops rotating, so it is now facing direction D

    Suppose a LiDAR scan is taken at each location, giving you scan C and scan D. Describe how you would relate the data points in scan D to the points in scan C. What is the added challenge for rotation compared to translation?

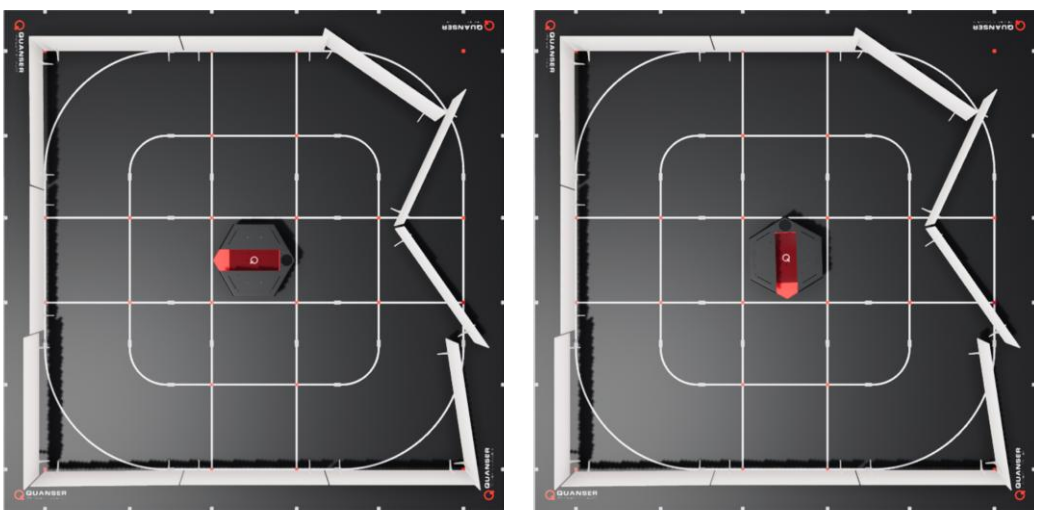

Figure 3. Top view of QBot facing direction C (left) and direction D (right)

Setting up LiDAR Scan Match

3. In this lab you will be exploring LiDAR-based localization using the Scan Match algorithm. In the Simulink model localization.slx, open the **LiDAR & Localization** subsystem. This subsystem will read data from the LiDAR sensor and perform scan match to estimate the position of the QBot. Open the **Subsample and Correct LiDAR** subsystem, then open the function block **correct_lidar** and complete the function with your code from Lab 4: Obstacle Detection in Skills Progression 1. Recall that this code corrects the raw headings so that 0 corresponds to the front of the QBot. If needed, review the lab procedure from Lab 4: Obstacle Detection.

4. Return to the **LiDAR & Localization** subsystem. Next, we will set up the inputs to the Scan Match block to estimate the position of the QBot using the subsampled LiDAR scan data. Open the **Run Scan Match Block** subsystem. This subsystem runs the **LIDAR Scan Match** block, which matches a reference LiDAR scan to the current LiDAR scan, performing a grid matching algorithm to determine the translation and rotation of the current scan relative to the reference. Right-click on this block and click on "help" to open the documentation. You can refer to this page for descriptions of input and output ports as we complete the rest of this lab.

5. Return to the **LiDAR & Localization** subsystem. In this lab, we will find the position of the QBot relative to its initial position, by using scan match to determine the transformation between its current scan and a saved reference scan from its initial position. First, connect the **correct_lidar** outputs to the **Run Scan Match Block** input ports for the current scan.

6. Next, we will complete the connections to the triggered **Freeze Scan** subsystem, which is set up to run with a rising edge trigger occurring when at least 3 seconds have elapsed. A triggered subsystem holds its outputs at the last value between triggers. Open the subsystem, where you can see that each input port connects directly to an output port. Return to the **Lidar & Localization** subsystem. Complete the connections to the **Freeze Scan** subsystem so it saves a copy of the current scan when triggered and passes the scan to the **Run Scan Match Block** subsystem.

7. The output of the **Lidar & Localization** subsystem is a data bus that contains current scan data, reference scan data, and the estimated pose output from the Scan Match Block. Complete the connections to the bus creator shown in Figure 4.

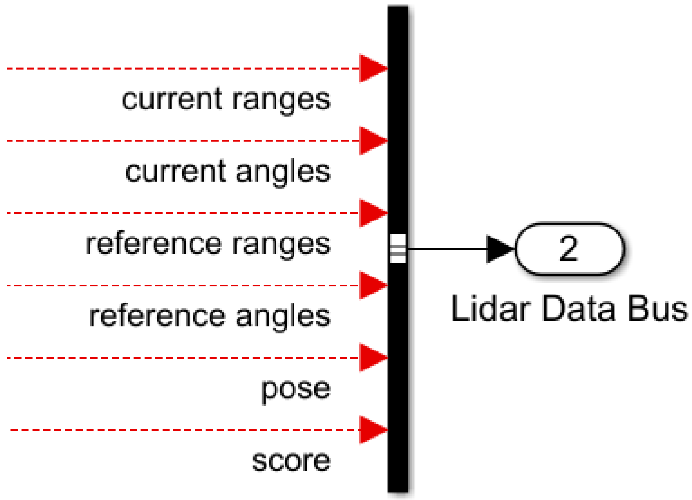

Figure 4: LiDAR Data Bus inputs

8. Return to the main Simulink model and note that the pose is selected from the Lidar Data Bus and then passed back to the **Lidar & Localization** subsystem. Open the Lidar & Localization subsystem again. The **LIDAR Scan Match** block starts the grid search at an initial position passed to the **initial** input. Connect this input to use the previous estimated position as the starting point for the current search.

9. The connections for the last two inputs are completed for you. The input **trans**, set to **[3, 3]**, specifies the extent of the translation search, in m, in the x and y directions. The input **rot** sets the angular search range in radians and is set to 2π.

10. Return to the main Simulink model. Click Monitor & Tune  in the Hardware or QUARC ribbon to deploy and run the model. When the model is run successfully, the QBot Platform LEDs will turn blue.

11. In the main Simulink model, open the **Data Display** subsystem. Open the Current Scan display, which displays the LiDAR scan data that is passed into the **currng** input of the **LIDAR Scan Match** block. Open the Current Pose display, which plots the current estimated position of the QBot relative to the reference LiDAR scan.

12. Drive the QBot around using the joystick controller by first pressing the left button (LB) to arm the QBot and then moving the left and right joysticks to control movement. Note that this lab does not include obstacle detection functionality. Observe the Current Scan and Current Pose displays as you drive the QBot around. Which heading corresponds to the forward movement direction of the QBot?

13. Open Data Scopes within the **Data Display** subsystem, which opens four different plots. Drive the QBot around while observing these scopes. The Estimated Pose plot displays the x, y, and theta values of the current position estimate. Which axis corresponds to the forward movement direction of the QBot? The Score plot displays

a non-normalized confidence for the match. The LiDAR Timing scope displays timing information for elements of the model related to the LiDAR sensor, including reading data and scan matching. The timing information for the rest of the model is displayed in the Main Timing scope. Press the right button (RB) to stop the code.

Exploring Parameters of LiDAR Scan Match

14. In this section, we will examine how scan match parameters impact performance. Open the LiDAR and Localization subsystem, then open the Run Scan Match Block subsystem.

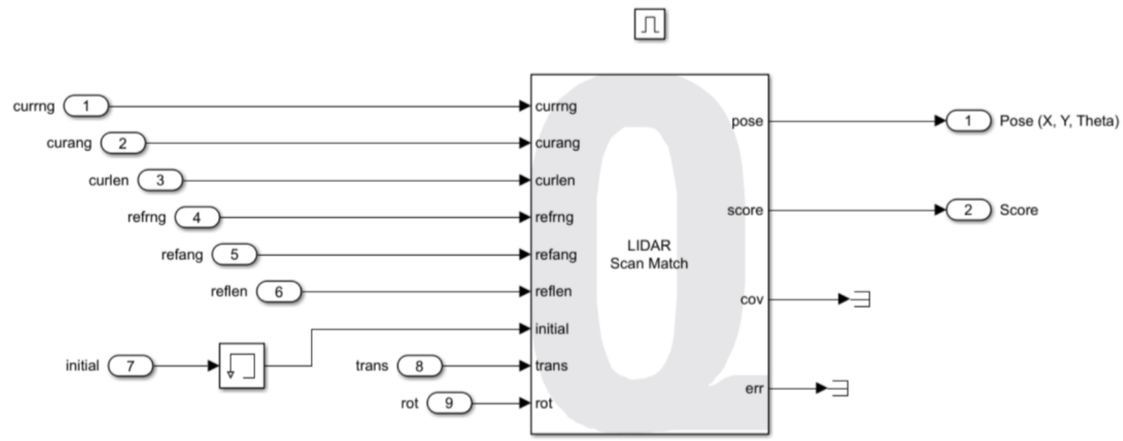

Figure 5. Inside of Run Scan Match Block subsystem

Click on the LIDAR Scan Match block, shown in Figure 7, to open block parameters. The **resolution** parameter, currently set to 20, sets the number of grid cells per meter. Increase the resolution to 40 and run the code. Drive the QBot around the map while observing the Estimated Pose, Score and Timing scopes. How does changing the resolution impact these values? If needed, increase the resolution by increments of 5 until you observe a significant effect. Stop your code.

15. Set the resolution back to 20. Inside the Lidar & Localization subsystem, change the constant connected to the **trans** input of the Scan Match Block to `[1 1]`. Run the code and drive the QBot around while observing the scopes. How is performance affected by changing the translation search range?

16. Stop the Simulink model when complete by pressing the right button (RB). Ensure that you save a copy of your completed files for review later.

17. Turn OFF the robot by single pressing the power button (do not keep it pressed until it turns off). Post shutdown, all the LEDs should be completely OFF.